MILITARY SPECIFICATION

COMPOUND, SEALING, FLUID RESISTANT

1. SCOPE

1.1 Scope. This specification covers two part synthetic rubber sealing compounds which are resistant to degradation by commonly used aircraft maintenance fluids.

1.2 Classification.

1.2.1 The sealing compound shall be of the following classes, as specified herein:

Classes:

Class A - Sealing and repair material suitable for brush application.

Class B - Sealing and repair material suitable for application by extrusion gun and spatula.

1.2.2 Dash numbers. The following dash numbers shall be used to designate the minimum application time in hours:

Class A dash numbers shall be: -½, -2.

Class B dash numbers shall be: -¼, -½, -2, -4, -6

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue contractually in effect, form part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-A-250/12	Aluminum Alloy, 7075, Plate and Sheet
QQ-A-250/13	Aluminum Alloy, Alclad 7075, Plate and Sheet
TT-E-751	Ethyl Acetate, Technical
TT-I-735	Isopropyl Alcohol
TT-M-261	Methyl-Ethyl-Ketone, Technical
TT-N-95	Naphtha, Aliphatic
TT-N-97	Naphtha, Aromatic
TT-S-735	Standard Test Fluids, Hydrocarbon
TT-T-548	Toluene, Technical

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CCC-C-419	Cloth, Cotton, Duck, Unbleached, Plied-Yarns Army and Numbered
PPP-C-96	Can, Metal, 28 Gage and Lighter

Military

MIL-S-5059	Steel, Corrosion, Resistant (18-8), Plate, Sheet and Strip
MIL-C-5541	Chemical Films and Chemical Film Materials For Aluminum and Aluminum Alloys
MIL-H-5606	Hydraulic Fluid, Petroleum Base, Aircraft, Missile and Ordnance
MIL-L-7808	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base
MIL-P-8585	Primer Coating, Zinc Chromate, Low-Moisture Sensitivity
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-S-8802	Sealing Compound, Temperature Resistant, Integral Fuel Tanks and Fuel Cell Cavities, High-Adhesion
MIL-T-9046	Titanium and Titanium Alloy, Sheet, Strip and Plate
MIL-R-25134	Remover, Paint and Lacquer, Solvent Type
MIL-C-25769	Cleaning Compound, Aircraft Surface, Alkaline Waterbase
MIL-C-27725	Coating, Corrosion Preventive, for Aircraft Integral Fuel Tanks
MIL-C-38334	Corrosion Removing Compound, Prepaint, for Aircraft Aluminum Surfaces

STANDARDS

Federal

FED-STD-601	Rubber; Sampling and Testing
FED-STD-791	Lubricants, Liquid Fuels, and Related Products; Method of Testing

Military

MIL-STD-129	Marking for Shipment and Storage

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

American Society for Testing and Materials (ASTM) Standards - Methods of Test for Mercaptan Sulfur in Aviation Turbine Fuels - ASTM-D-1323
Identation Hardness of Rubber and Plastic by Means of a Durometer, Test For - ASTM D-2240

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TABLE I - APPLICATION, TACK-FREE, AND CURE TIMES

CLASS	APPLICATION TIME (HOURS)	TACK-FREE TIME (HOURS)	CURE TIME (HOURS)
B-¼	¼	1	8
A or B-½	½	8	16
A or B-2	2	24	48
B-4	4	36	72
B-6	6	48	168

3.4.6 Hardness. When tested in accordance with 4.9.7, the sealing compound shall have a hardness between 60 and 70 after curing under standard conditions (see 4.6.3) for not more than the time specified in Table I.

3.4.7 Elevated temperature resistance. When tested in accordance with 4.9.8, the sealing compound shall not blister or sponge, and shall retain a minimum of 150 psi tensile strength after 200°F exposure. The durometer hardness and elongation shall not change more than 20 percent over initial values.

3.4.8 Low-temperature flexibility. When tested in accordance with 4.9.9, the sealing compound shall withstand the low-temperature flexibility test at minus 65 ± 2°F without cracking, checking, or loss of adhesion.

3.4.9 Adhesion.

3.4.9.1 Peel strength. When tested in accordance with 4.9.10, the cured sealing compound shall have a minimum initial peel strength of 10 pounds per inch and shall exhibit 100 percent cohesive failure except for bubbles, knife cuts, and other causes obviously not the fault of the sealant.

3.4.9.2 Lap shear strength. When tested in accordance with 4.9.10.1, the cured sealant shall have a minimum lap shear strength of 200 psi. It shall exhibit 100 percent cohesive failure, except for bubbles or other causes that are obviously not the fault of the sealing compound.

3.4.9.3 Adhesion to MIL-S-8802. When tested in accordance with 4.9.10.2, the sealing compound shall have a minimum peel strength of 10 pounds and shall not separate between the MIL-S-8802 and the sealant being tested.

3.4.10 Corrosion. When tested in accordance with 4.9.11, the sealing compound shall cause no visible corrosion of the aluminum panels and shall show no sign of deterioration.

3.4.11 Resistance to maintenance fluids.

3.4.11.1 Weight loss and flexibility. When tested in accordance with 4.9.12.1, the sealing compound shall not lose more than 8½ percent by weight. The specimens shall not crack when bent 180 degrees over an 1/8-inch mandrel.

3.4.11.2 Tensile strength and elongation. The Class B compound, when tested in accordance with 4.9.12.2, shall have an ultimate tensile strength and elongation of the cured compound not less than the values shown in Table II.

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Qualification. The sealing compound furnished under this specification shall be a product which has passed the qualification tests specified herein and has been listed or approved for listing, on the applicable Qualified Products List. When a product has been qualified, any change in the constituent materials or method of manufacture shall require requalification.

3.2 Materials. The basic ingredient used in the manufacture of the sealing compound shall be synthetic rubber which may be of the polyurethane or other type. The sealing compound shall cure by the addition of a separate curing compound to the base compound and shall not depend on solvent evaporation for curing. Manufacturers will be given wide latitude in the selection of materials and curing systems. Special cleaners or primers may be used and are authorized to insure adhesion.

3.3 Appearance. The base compound and the curing compound (accelerator) shall each be of uniform blend and shall be free of skins, lumps, and jelled or coarse particles. There shall be no separation of ingredients which cannot be readily dispersed.

3.4 Physical properties.

3.4.1 Specific gravity. When tested in accordance with 4.9.1, the specific gravity of the cured sealing compound shall not exceed 1.60 at standard conditions (see 4.6.3).

3.4.2 Non-volatile content. When tested in accordance with 4.9.2, the non-volatile content of the freshly mixed compound shall be not less than the amount specified below:

Class A - 92 percent by weight
Class B - 98 percent by weight

3.4.3 Viscosity of mixed sealant. Class A sealing compound shall be suitable for application by brush. The viscosity of freshly mixed Class A sealing compound, except Class A ¼, shall be within the range of 100 to 500 poises when tested in accordance with 4.9.3. Class B sealing compound shall be suitable for application by extrusion gun or spatula. The viscosity of freshly mixed Class B sealing compound, except Class B ¼, shall be within the range of 6,000 to 16,000 poises when tested as specified in 4.9.3.

3.4.4 Flow (Class B only). When tested in accordance with 4.9.4, freshly mixed Class B sealing compound shall exhibit an initial flow within the limits of 0.1 to 1.5 inches when a cylindrical section formed in the flow-test jig is allowed to flow under its own weight in a vertical surface. In addition, the sealing compound shall retain this flow within the application time.

3.4.5 Application, tack-free, and cure-times. The sealing compound shall conform to the application, tack-free, and cure times specified in Table I, when tested as specified in 4.9.5, 4.9.6 and 4.9.7.

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TABLE II - TENSILE STRENGTH AND ELONGATION

CONDITION	TENSILE STRENGTH PSI MIN	ELONGATION (%) MIN
Full cure (unimmersed) tested at room temperature	150	80
7 days at 200°F tested at RT	150	80
8 hrs immersion in cleaner, MIL-C-25769	150	80
8 hrs immersion in corrosion removing compound, MIL-C-38334	150	80
8 hrs immersion in Turco Fabrifilm Remover**	150	80
8 hrs immersion in acid-type paint stripper (B & B 1717 AMX)*	120	80
8 hrs immersion in paint stripper MIL-R-25134	120	80
8 hrs immersion in phenol type paint stripper (Turco 5351)**	120	80
Following at 140°F		
7 days immersion in water	150	80
7 days immersion in jet reference fluid	150	80
7 days immersion in engine oil, MIL-L-7808	100	80
7 days immersion in hydraulic oil, MIL-H-5606	150	80

*B & B Chemical Company
**Turco Products, Inc

3.4.11.3 Lap shear strength. The cured sealant, when tested per 4.9.12.3, shall have shear strengths not less than those shown in Table III after fluid immersion.

TABLE III - SHEAR STRENGTH

CONDITION	SHEAR STRENGTH PSI
After 8 hrs immersion in cleaner MIL-C-25769	150
After 8 hrs immersion in brightener MIL-C-38334	150
After 8 hrs immersion in acid-type paint stripper (B&B 1717 AMX)*	120
After 8 hrs immersion in paint stripper, MIL-R-25134	120
After 8 hrs immersion in Phenol type paint stripper (Turco 5351)**	120
Following at 140°F	
After 7 days immersion in water	150
After 7 days immersion in jet reference fluid	150
After 7 days immersion in engine oil MIL-L-7808	150
After 7 days immersion in hydraulic oil MIL-H-5606	150

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3.4.12 Accelerated storage stability. After storing in accordance with 4.9.13, neither the base compound nor the curing agent shall show skinning, hardening, separation, nor settling of the material. The mixed material shall also conform to the viscosity specified in 3.4.3. The aged material, when mixed, shall conform to the requirements for flow, application time, and tack-free time and cure time specified in 3.4.5.

3.4.13 Repairability. The Class A and Class B sealing compounds shall be suitable for repairing minor breaks in themselves or each other and in all other aged sealing compounds qualified to this specification. Specimens of sealing compound prepared in accordance with 4.9.14 shall have a minimum peel strength of 10 pounds per inch.

3.4.14 Hydrolytic stability. After exposure to the conditions specified in 4.9.15, the cured material shall not decrease in Durometer Shore A hardness more than 30% of the initial value prior to exposure on any specimen.

3.4.15 Color. The accelerator used shall possess sufficient color contrast to the base compound to permit easy identification of an unmixed or incompletely mixed sealing compound. The color of the cured compound shall be gray to silvery gray.

3.5 Workmanship.

3.5.1 The workmanship shall be in accordance with high grade manufacturing practice covering this type of material. It shall be suitable for its intended use and free of defects which may affect its performance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests.

4.2.1 The inspection and testing requirements specified herein shall be classified as follows.

4.2.1.1 Qualification tests. Those tests and inspection requirements which are necessary to establish that the material or product conforms to all specified requirements.

4.2.1.2 Lot acceptance tests. Those tests to be performed on individual items or lots of items to determine that certain specified requirements, which serve as a basis for acceptance, have been met.

4.3 Conformance certification.

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4.6.2 Lot acceptance tests. A one-quart sample, representative of each lot of sealing compound on order, shall be selected from each lot and subjected to the tests listed below.

a. Non-volatile content.

b. Viscosity of mixed sealant.

c. Flow.

d. Application time.

e. Tack-free time.

f. Adhesion (peel test unimmersed).

4.6.3 Test conditions. Standard laboratory testing conditions shall be 77 ± 2°F and 50 ± 5 percent relative humidity. Except as otherwise specified herein, all test specimens shall be prepared, cured, and tested under these conditions. Specimens for performance tests may be cured for 24 hours at 130 ± 2°F to expedite laboratory testing.

4.7 Preparation of test specimens.

4.7.1 Cleaning of test panels. All test panels shall be cleaned by scrubbing and rinsing with soap-free and grease-free rags or paper towels, using solvent formulated in accordance with Table IV. Immediately after rinsing, the panels shall be wiped dry with a clean, lint-free cloth. The use of special cleaners or primers to insure adhesion is permitted. Apply the cleaner or primer to panels in accordance with the supplier's recommendations.

TABLE IV - CLEANING SOLVENT INGREDIENTS

INGREDIENT	SPECIFICATION	PERCENT BY VOLUME
Aromatic petroleum naphtha	TT-N-97, Type I, Grade B	50
Ethyl acetate	TT-E-751	20
Methyl-ethyl-ketone	TT-M-261	20
Isopropyl Alcohol	TT-I-735, Grade B	10

4.7.2 Preparation of sealing compound. The quantity of sealing compound required for the test shall be mixed as thoroughly as practicable. A Semco Model 1394 or Model SP-1350 mixer, or equal, shall be used. The mixed sealing compound shall have a minimum inclusion of air. Where applicable, the sealing compound, immediately after mixing, shall be placed into cartridges for extrusion from the Semco No. 250 gun, or equal.

4.7.3 Application of sealing compound. Unless otherwise specified herein, test panels shall be given an application of sealing compound to produce a coating having a thickness of 1/8 ± 1/64 inch when cured. Class A material shall be built-up by three applications allowing 2 hours between applications to permit release of solvents. Sealants shall be applied on properly primed

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4.3.1 Qualification test report. The supplier shall furnish to the Government two copies of a certified report of the results of the qualification tests specified herein, performed to determine conformance with the requirements of this specification. The Qualification Test Report shall be submitted upon completion of the tests (see 6.3).

4.3.2 Acceptance test report. Upon request, the supplier shall furnish to the Government two copies of a certified report of the results of the acceptance tests specified herein, performed to determine conformance to the requirements of this specification.

4.3.3 Certificate of conformance. Upon request, the supplier shall furnish two copies of a certificate of conformance confirming that all material or products in the shipment meet and comply with the requirements of this specification. The absence of such a request shall in no way relieve the supplier of the responsibility for furnishing materials or products which meet all the requirements specified herein, and supporting data for such a certificate shall be retained by the supplier and shall be available to the Government at all times when duly requested.

4.3.4 Conformance certification data. Test reports and certificates of conformance furnished in accordance with these requirements shall include the following identifying information, where applicable:

a. Purchase order number.

b. Material specification number, revision, and class.

c. Date of manufacture.

d. Manufacturer's stock or lot designation.

e. Date and description of tests performed.

4.4 Qualification tests. Qualification tests for sealing compound to be furnished under this specification shall consist of the tests specified in 4.6 through 4.9.15.

4.5 Lot acceptance tests. Lot acceptance tests for sealing compound furnished under this specification shall consist of the tests specified in 4.6.2.

4.5.1 Lot. Unless otherwise specified, a lot shall consist of all sealing compound manufactured at one time from one batch, forming part of one contract or order for delivery.

4.6 Qualification tests.

4.6.1 Qualification test samples. The qualification test samples shall consist of 3 quarts of compound upon which qualification is desired. The formulation for the Class A and other Class B materials must be the same as that for the original Class B sealant qualified, except for minor variations necessary for viscosity and application time.

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surfaces or panels properly prepared with cleaner/primer when required. Applied sealant shall be fully cured prior to being subjected to any tests.

4.8 Formulation of jet reference test fluid. The jet reference fluid required for conducting fluid immersion tests shall be formulated as follows:

a.	Toluene (TT-T-548)	30 Volumes
b.	Cyclohexane (Technical Grade)	60 Volumes
c.	Iso-Octane (TT-S-735, Type I)	10 Volumes
d.	*Tertiary dibutyl disulfide	1 Volume
e.	Tertiary butyl mercaptan	0.015 ± 0.0015 weight percent of other four components

*The tertiary dibutyl disulfide shall be doctor sweet.

4.8.1 When tested for mercaptan and total sulfur in accordance with ASTM D 1323 and Method 5201 of FED-STD-791, respectively, the reference fluid shall have the following properties:

a.	Total Sulfur Content	0.400 ± 0.005 weight percent
b.	Mercaptan Sulfur Content	0.0050 ± 0.0005 weight percent

The fluid shall be stored out of contact with light in containers which are inert to the fluid ingredients. (Welded aluminum, non-galvanized welded steel, or glass containers are suitable.) If not used within 90 days after blending, the fluid should be retested for mercaptan and total sulfur content.

4.9 Test methods.

4.9.1 Specific gravity. Three specimens approximately 1/8 inch by 1 inch by 1 inch in size shall be cut with a sharp razor blade from a sheet of sealing compound that has been given a full cure (see 4.6.3). The specimens shall be weighed accurately to three decimal places in grams after which they shall be dipped in methyl alcohol and then, while wet, immediately suspended in distilled water at standard conditions and reweighed. The specific gravity shall be computed from the following formula:

$$\text{Specific Gravity} = \frac{\text{Weight in Air}}{\text{Weight in Air - Weight in Water}}$$

4.9.2 Non-volatile content. Five to 10 grams of freshly mixed sealing compound shall be transferred as rapidly as possible to a cup approximately 3 inches in diameter and 3/4 inches in depth. A fitted cover shall immediately be placed over the cup and the weight determined accurately, using an enclosed beam balance which will weigh accurately to the nearest milligram. The weight of the cup and cover shall be determined accurately prior to using, and subtracted from the initial and final weights in order to calculate the new sample weights. The cover shall then be removed, and the sealing compound heated in an oven for 7 days at 158° ± 5°F. It shall then be cooled in a desiccator, the

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cover replaced, and the weight determined accurately. The percentage of total non-volatile content shall be calculated as follows:

$$\text{Percent non-volatile content} = \frac{\text{Final Weight X 100}}{\text{Initial Weight}}$$

4.9.3 Viscosity of mixed sealant. The sealing compound shall be stored at room temperature for 8 hours minimum prior to test. The freshly mixed sealing compound shall be placed in quart cans and the viscosity of the mixed sealant measured within 5 minutes after mixing. The temperature of the freshly mixed sealant at time of test shall be 77 + 2°F. The Brookfield Model RVF viscosimeter, or equal, shall be used and the readings obtained converted to poises. For Class A sealant, the No. 4 or No. 5 spindle at 10 revolutions per minute shall be used; for Class B sealant, the No. 7 spindle at 2 revolutions per minute shall be used. The highest reading shall be taken from each sample after the instrument has run in the material for 1 minute.

4.9.4 Flow. The Class B base compound, accelerator, and sealant application gun shall be stabilized at standard conditions for at least 8 hours before a minimum 250-gram lot of the base compound is mixed with a proper amount of accelerator. The mixed sealing compound shall be promptly used to fill a standard Semco sealant gun cartridge, or equal, having a Semco 250 N-4-4 nozzle or equal. The gun and sealing compound shall be maintained at standard conditions throughout the test. The test shall be conducted with a flow-test jig as shown in Figure 1 and under flow conditions specified in 3.4.4. Depth of plunger tolerance is critical and shall be controlled within the tolerances during all tests.

4.9.4.1 The flow-test jig shall be placed on a table with the front facing upward and the plunger depressed to the limit of its travel. Within 15 minutes after the beginning of mixing, enough of the mixed sealing compound shall be extruded from the application gun to fill the recessed cavity of the jig and leveled off even with the block. Within 10 seconds after the leveling operation, the jig shall be placed on its end and the plunger immediately advanced to the limit of its forward travel. The flow shall be measured from tangent to the lower edge of the plunger to the farthest point to which flow has advanced.

4.9.4.2 As the sealing compound progresses in its application time, the flow test shall be repeated at the time intervals specified in Table V.

TABLE V - FLOW-TEST INTERVALS

MINIMUM SPECIFIED APPLICATION TIME	INTERVALS AT WHICH FLOW-TESTS SHALL BE CONDUCTED
1/4 and 1/2 hour	Initial
2 hours	Initial, 1 hour, 1-1/2 hours
4 hours	Initial, 2 hours, 3 hours
6 hours	Initial, 3 hours, 5 hours

4.9.5 Application time.

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4.9.9 Low-temperature flexibility. Four 0.040 by 2-3/4 by 6 inch aluminum alloy panels conforming to temper T6 of QQ-A-250/13 shall be sulfuric acid anodized with a dichromate seal in accordance with MIL-A-8625 and a 1 mil coating (approx) of MIL-C-27725 applied and cured. Apply one coat of sealant primer, if required. A coating of the sealing compound 0.094 inch by 1-1/2 inches by 4 inches shall be applied to the center of each of the 4 panels. Care shall be taken to maintain an accurate sample thickness of 0.094 inch. Panels shall be cured in accordance with 3.4.5. All four test panels shall then be placed in a low temperature flexibility jig in accordance with Figure 2. The jig shall be capable of flexing the panel through a 30-degree arc (15 degrees each side of the center) at a constant speed of 1 cycle per 5 seconds. The temperature shall be reduced to -65 ± 2°F, stabilized at this temperature for 2 hours, and the panels flexed through 130 consecutive cycles. Conformance to 3.4.8 shall be demonstrated.

4.9.10 Peel strength. Six 0.040 inch by 3-3/4 inches by 6 inches QQ-A-250/13 T6 aluminum alloy test panels shall be prepared. Two panels shall be sulfuric anodized with a dichromate seal in accordance with MIL-A-8625 and a 1 mil coating (approximate) of MIL-C-27725 applied and cured. Two panels shall be conversion coated and a 1/2 mil coating (approximate) of MIL-P-8585 applied and cured. An additional coat of sealant primer may be applied, if required. Two panels shall be color conversion coated in accordance with MIL-C-5541 and primed with the sealant primer (if required). All six panels shall be coated with a 1/8 ± 1/64 inch thickness of sealing compound. After the sealing compound has immobilized, one end of a 3 by 12 inch strip of wire screen (20 to 40 mesh aluminum or monel - wire fabric) or cotton duct conforming to Type III of CCC-C-419, shall be placed on each panel, leaving a loose end 6 inches in length. The wire fabric shall be coated with primer, when required, to fill all holes. An additional 1/32 inch thick coating of sealing compound shall be applied over the screen or fabric strips and the panel allowed to cure for the approximate cure time specified in 3.4.5, Table I. Peel tests shall be conducted on the panels as follows: A 1-inch wide section shall be cut through the strip and sealing compound on each test panel to provide the center inch for the peel test. The specimens shall be stripped back at an angle of 180 degrees to the metal panel in a suitable tensile testing machine having a jaw separation rate of 2-inches per minute. The results shall be the numerical average of the peak loads. Conformance to 3.4.9.1 shall be demonstrated.

4.9.10.1 Lap shear strength. Twenty lap-shear specimens shall be prepared. Twenty-four strips 1 by 5 by 0.051 inch made from aluminum alloy conforming to QQ-A-250/13, shall be solvent cleaned and dried. Eight strips shall be coated with approximately 1/2 mil MIL-P-8585, Zinc Chromate Primer, and 8 strips with 1 mil MIL-C-27725 polyurethane coating; and 8 left bare. The coating shall be fully cured. Eight titanium panels conforming to MIL-T-9046 and eight steel panels conforming to MIL-S-5059 shall be solvent cleaned and dried. The test panels may be primed with sealant primer if one is required, then freshly mixed sealant shall be applied to one square inch at one end of each panel to provide a sealant thickness of approximately 0.010 inch. The test strips shall then be jig assembled to provide 20 lap-shear specimens having a one square inch overlap. Sufficient pressure shall be maintained to provide a sealant thickness in the faying surfaces of 0.010 ± 0.004 inch. After a full cure, as specified in Table I, the specimens shall be pulled in a tensile machine operating with a jaw separation of 2-inches per minute. Conformance to 3.4.9.2 shall be demonstrated.

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4.9.5.1 Class A sealing compound. Both components of the sealing compound shall be stabilized at standard conditions for a minimum of 8 hours before mixing in the proper ratios as specified by the manufacturer. A sufficient quantity shall be mixed to fill a standard 1/2 pint can (2-7/8 inches in diameter by 2-7/8 inches high) to within 1/2 inch of the top. This can shall be tightly covered except when testing the viscosity. At the end of the rated application time (see 3.4.5), measured from the beginning of the mixing period, the sealing compound shall be tested for viscosity using a Brookfield Model RVF viscosimeter, or equal, with a No. 7 spindle at 10 rpm. One reading shall be taken after the instrument has run in the material for 1 minute. The viscosity of the mixed material shall not exceed 2500 poises at the end of the rated application time.

4.9.5.2 Class B sealing compound. Both components of the sealant and application gun shall be stabilized at standard conditions for at least 8 hours before mixing, and 250 grams of the sealant shall be properly mixed and used to fill a standard Semco sealing gun cartridge, or equal, having a Semco 440 nozzle with an orifice chamber of 0.125 ± 0.005 inch. The gun and sealing compound shall be maintained at standard conditions throughout the test. The gun shall be attached to a constant air supply of 90 ± 5 psig. From 2 to 3 inches of sealing compound shall be extruded initially to clear trapped air. At the end of the rated application time, measured from the beginning of the mixing period, the sealing compound shall be extruded onto a suitable receptacle for 1 minute and the weight of extruded sealing compound determined. The weight of extruded sealant shall not be less than 15 grams at the end of the rated application time.

4.9.6 Tack-free time. An aluminum test panel conforming to temper T6 of QQ-A-250/13 and measuring 0.040 inch by 2-3/4 inches by 6 inches in size shall be cleaned in accordance with 4.7.1 and covered with freshly mixed sealing compound to a depth of 1/8 ± 1/64 inch. The sealing compound shall be allowed to cure at standard conditions and at the rate specified in 3.4.5. At the end of the specified tack-free time, two 1-inch by 6-inches pieces of polyethylene film 0.004 ± 0.002 inch thick shall be applied to the sealing compound and held in place at a pressure of 1/2-ounce per square inch for 2 minutes. The strip shall then be slowly and evenly withdrawn at right angles to the sealing compound surface. The polyethylene shall come away clean and free of sealing compound.

4.9.7 Cure time and hardness. The instantaneous hardness shall be determined in accordance with Method 3021 of FED-STD-601 after the sealing compound is allowed to cure as specified in 3.4.5. The reading shall be taken on a doubled back-to-back 0.125-inch thick specimen.

4.9.8 Elevated temperature resistance. Tensile test specimens shall be prepared and cured in accordance with 4.9.12.2. Test specimens shall be subjected to 200°F dry heat for 8 hours. The specimens shall be cooled to room temperature and the hardness, tensile strength, and elongation shall be determined and compared with initial values (controls). The test shall be conducted in accordance with Method 4111 of FED-STD-601. This test shall be conducted not less than 16 hours nor more than 48 hours after removal from oven. The instantaneous hardness of the aged specimen shall be determined in accordance with Method 3021 of FED-STD-601.

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4.9.10.2 Adhesion to MIL-S-8802. Three 0.040 inch by 2-3/4 inch by 6 inch test panels conforming to temper T6 of QQ-A-250/13 shall be cleaned in accordance with 4.7.1 and coated with a 1/8 ± 1/64 inch thickness of MIL-S-8802, Class B sealant. The sealant shall be cured in accordance with MIL-S-8802. A sealant primer may be used, if required, for adhesion of the sealant to be tested. A 1/8 ± 1/64 inch thick coat of sealant to be tested shall be applied. After the sealing compound has immobilized, one end of a 3 inch by 12 inch strip of wire screen, 20 to 40 mesh aluminum or monel wire fabric or CCC-C-419, cotton duck, Type III shall be placed on each panel, leaving a loose end 6 inches in length. The wire fabric shall be coated with primer, when required, to fill all holes. An additional 1/32 inch thick coating of the sealing compound to be tested shall be applied over the screen or fabric strips. The strips shall be allowed to cure for the time specified in Table I. A 1-inch wide section shall be cut through the fabric or screen and sealing compound on each test panel to provide the center inch for the peel test. The specimen shall be stripped back at an angle of 180 degrees to the metal panel in a suitable tensile testing machine having a jaw separation rate of 2 inches per minute. The results shall be the numerical average of the peak loads. Conformance to 3.4.9.3 shall be demonstrated.

4.9.11 Corrosion. One aluminum panel conforming to QQ-A-250/12 and measuring 0.040 by 2-3/4 by 6 inches in size shall be prepared with two parallel sealing compound fillets (using sealant primer, if required) approximately 1-inch apart. The fillets shall be applied 3/16 inch thick by 3/4 inch wide by 5 inches long and shall extend to within 1/2 inch of the edges of the panel. The sealant shall be given a full cure, as specified in Table I. After sealant has cured, the panel (and fillets) shall be completely overcoated with MIL-C-27725 approximately 1 mil thick. The coating material shall be given a full cure. The fillets shall be cut down to 1/16 inch and panels immersed vertically for 20 days in a covered glass vessel containing a 3 percent aqueous sodium chloride solution at 140 ± 2°F. Immediately upon removal from the fluid, the panel shall be examined for compliance with the requirements of 3.4.10.

4.9.12 Resistance to maintenance fluids.

4.9.12.1 Weight loss and flexibility. Twelve 1/8 by 1 by 5 inch specimens shall be cut from a sheet of sealing compound that has been fully cured, as specified in Table I. Four specimens shall be weighed and immersed in 900 cubic centimeters of each of the following fluids at 140 ± 2°F in a closed container for a period of 7 days:

a. Jet reference fluid, as specified in 4.8.

b. Engine oil, MIL-L-7808.

c. Hydraulic oil, MIL-H-5606.

At the end of the exposure period, the specimens shall be removed from the test fluids, flushed clean with TT-N-95, Aliphatic Naphtha, and air dried for 72 hours at 120 ± 2°F. The specimens shall then be cooled to standard test temperature in a desiccator and weighed. The average percent weight loss shall be calculated. After weighing, the specimens shall be bent 180 degrees over 1/8-inch mandrel and examined for evidence of cracking. Conformance to 3.4.11.1 shall be demonstrated.

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4.9.12.2 Tensile strength and elongation. Mixed sealing compound 0.060 ± 0.015 inch thick shall be prepared by pressing between two polyethylene sheets, removing the top sheet at the end of the rated tack-free time, and allowing the sealing compound to cure for the specified cure time in Table I. Thirty tensile test specimens shall be cut from the sheet, after full cure, using die VI as specified in Method 4111 of FED-STD-601. Three specimens shall be exposed in each of the following fluids for 8 hours at room temperature.

Cleaner, MIL-C-25769

Brightener, MIL-C-38334

Acid-Type Paint Stripper, Commercial (B & B 1717 AMX)

Paint Stripper, MIL-R-25134

Phenol Type Paint Stripper (Turco 5351)

Three specimens shall be immersed in each of the following fluids for 7 days at 140°F:

Water

MIL-L-7808 Engine Oil

MIL-L-5606 Hydraulic Oil

Jet Reference Fluid

After immersion, specimens shall be thoroughly rinsed in clean tap water. In the case of hydraulic fluid, and jet reference fluid, and engine oil clean items with TT-N-95 Aliphatic Naphtha and then dry for a minimum of 24 hours at room temperature prior to testing. Conformance to the requirements of Table II shall be demonstrated.

4.9.12.2.1 Test procedure. The tensile and elongation tests shall be conducted at room temperature in accordance with Methods 4111 and 4121, respectively of FED-STD-601.

4.9.12.3 Lap-shear strength. Twenty-seven lap-shear specimens shall be prepared. Controls for these tests shall be those items tested under paragraph 4.9.10.1. Fifty-four test strips, 1 by 3 by 0.051 inch, made from aluminum alloy conforming to QQ-A-250/13 shall be solvent cleaned with TT-N-95 Aliphatic Naphtha and dried. The test panels may be primed with a sealant primer, if one is required. The freshly mixed sealing compound shall be applied to a 1-inch square area at one end of each panel to provide a sealant thickness of approximately 0.010 inch. The test strips shall be jig assembled to provide 27 lap shear specimens having a one square inch overlap. Sufficient pressure shall be applied to specimens to provide a sealant thickness in the faying surfaces of 0.010 ± 0.004 inch. After full cure, per requirements of Table I, the lap joints shall be immersed in the following test fluids for the times specified:

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3 panels 8 hours in cleaner, MIL-C-25769

3 panels 8 hours in corrosion removing compound, MIL-C-38334

3 panels 8 hours in paint stripper, MIL-R-25134

3 panels 8 hours in paint stripper, B & B 1717 AMX

3 panels 8 hours in phenol type paint stripper (Turco 5351)

Three panels shall be immersed in each of the following fluids for seven days at 140°F:

Water

MIL-L-7808 Engine Oil

MIL-L-5606 Hydraulic Oil

Jet Reference Fluid

After immersion, the test panels shall be thoroughly rinsed clean in tap water (in the case of oil, hydraulic fluid, and jet reference fluid immersion tests, clean items in TT-N-95 Aliphatic Naptha) and then dry for a minimum of 24 hours at room temperature prior to testing. The specimens shall be pulled in a tensile machine operating with a jaw separation rate of two inches per minute. Conformance to requirements in Table III shall be demonstrated.

4.9.13 Accelerated storage. An original, unopened, 1-quart container of base compound and an original, unopened container of the curing compound shall be stored for 14 days at 120 ± 2°F in a suitable vented oven. After cooling at standard conditions for at least 24 hours, tests shall be conducted per 4.9.3, 4.9.4, 4.9.5, and 4.9.6. After storage, the base compound and catalyst shall be in accordance with 3.4.12.

4.9.14 Repairability. Two aluminum alloy panels 6 by 3 by 0.40 inch shall be cleaned, primed with sealant primer, if required, and coated on one side with sealant to provide a uniform thickness of approximately 1/16 inch. Sealant shall be fully cured per requirements of Table I. Panels shall be exposed to a weather-o-meter for 7 days. A cycle of 3 minutes water spray and 17 minutes sunshine shall be maintained. Exposure shall be followed by 3 days air-drying at 120 ± 2°F. The panels shall then be cleaned in accordance with 4.7.1 and the newly mixed sealing compound shall be applied over the existing sealing compound. A 3 by 12-inch strip of wire screen (20 to 40 mesh aluminum or monel wire fabric) or cotton duck conforming to Type III of CCC-C-419 shall be placed on each panel and approximately 1/32 inch of new sealing compound applied over the strip. After a full cure, as specified in Table I, the specimens shall be tested for peel, as specified in 4.9.10 and shall meet the requirements of 3.4.13.

4.9.15 Hydrolytic stability.

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4.9.15.1 Specimen preparation. Sufficient base compound and curing agent shall be mixed to prepare 5 molded test specimens, 2-1/2 inches in diameter by 1/2 inch thick. Each specimen shall be cured in accordance with Table I. Hardness shall be determined using a Type A Durometer in accordance with ASTM D2240 after 3 second application time. Hardness shall be determined at the same locations before and after exposure.

4.9.15.2 Procedure. After determining hardness before exposure, the specimens shall be placed vertically in a suitable holder, on a tray in a suitable glass desiccator (250 mm). The desiccator shall contain a glycerine (22 percent by weight) in water solution in the bottom which will produce a relative humidity (RH) of 95 percent at the test temperature. The desiccator, containing the specimens, shall then be closed and inserted into an air circulating oven maintained at 71 ± 1°C (160 ± 3°F) for a period of 120 days. At the end of the exposure period, the desiccator shall be removed from the oven and cooled to standard conditions (4.6.3) for 24 to 36 hours. Hardness shall be determined as specified in 4.9.15.1 and the obtained values for each specimen shall be in accordance with paragraph 3.4.14.

4.10 Packaging, packing and marking. Preparation for delivery shall be examined for compliance with section 5.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packaging. The base compound and the curing agent shall each be packaged in its own container. Each curing agent container shall be packaged with one base compound container in a manner that will prevent accidental separation but will permit easy separation for mixing purposes. The ratio of the quantity contained in base container to the quantity contained in the curing agent container shall be the same as the recommended mixing ratio of the base compound and curing agent.

5.1.1.1.1 Container size. The sealant shall be furnished in quantities, container size and type as specified in the contract or order.

5.1.2 Level C. Packaging shall be in accordance with the manufacturer's commercial practice.

5.2 Packing. Packing shall be Level A, B or C as specified (see 6.2).

5.2.1 Level A. The sealing compound shall be packed in overseas-type shipping containers in accordance with the requirements of the appendix to PPP-C-96.

5.2.2 Level B. The sealing compound shall be packed in domestic-type shipping containers in accordance with the requirements of the appendix to PPP-C-96.

5.2.3 Level C. The sealing compound shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation.

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5.3 Marking of shipment. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. Marking shall include specification number, applicable class and dash number, name of manufacturer, date of manufacture (month and year), batch number, mixing instructions, and recommended storage temperature.

6. NOTES

6.1 Intended use. The sealing compound covered by this specification is intended as a fillet type seal for sealing aircraft structures that may be subjected to a service temperature range of -65°F to 180°F, with limited use up to 200°F maximum. The sealing compound is not intended for integral fuel tanks; however, these compounds do resist fuel, hydraulic fluid and lubricating oil and are resistant to chemical paint strippers.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Class of sealant required and dash number (see 1.2).

c. Mixing instructions (parts per hundred by weight, curing agent to base).

d. Application time.

e. Quantity and size desired.

f. Applicable levels of packaging and packing required (see 5.1 and 5.2).

g. Any special marking required (see 5.3).

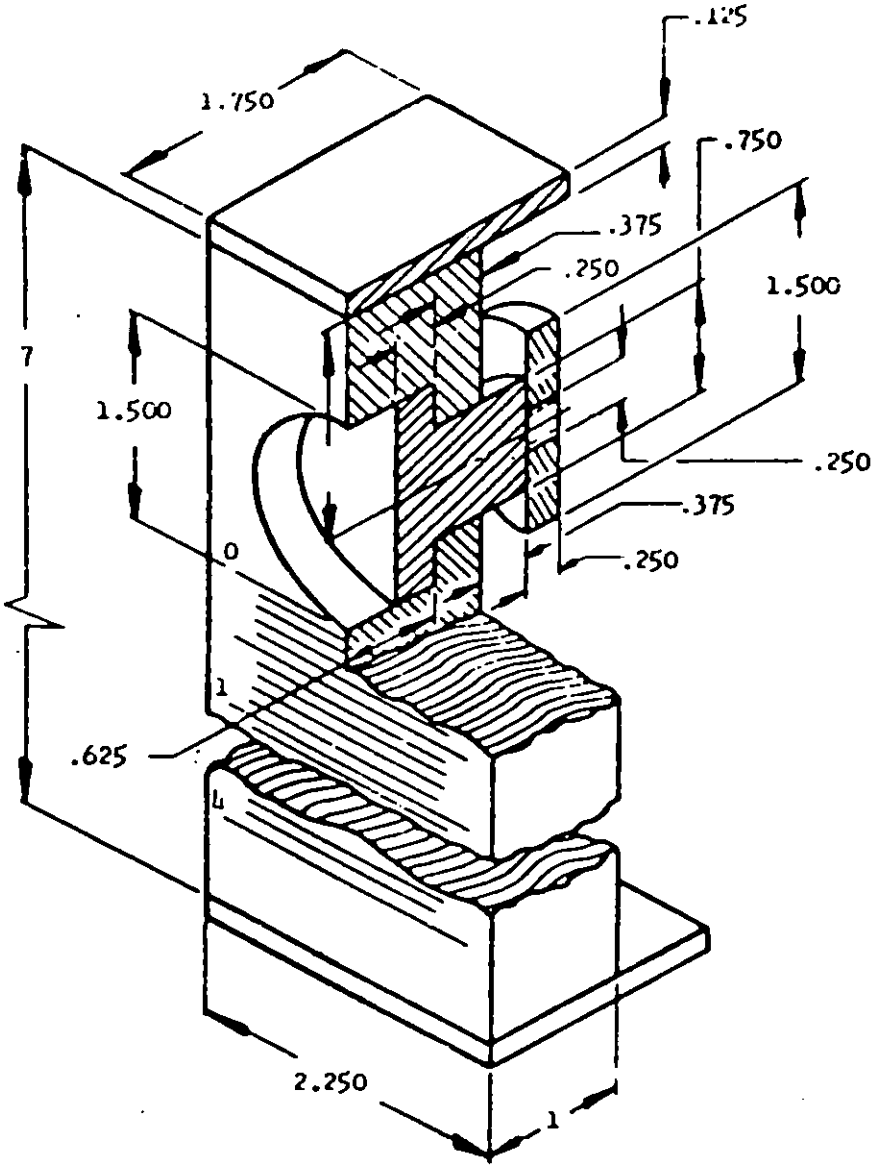
6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of suppliers is called to this requirement, and manufacturers are urged to have the products that they propose to offer the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is WRAMA/MMETM, Robins AFB GA 31098, and information pertaining to qualification of products may be obtained from that activity.

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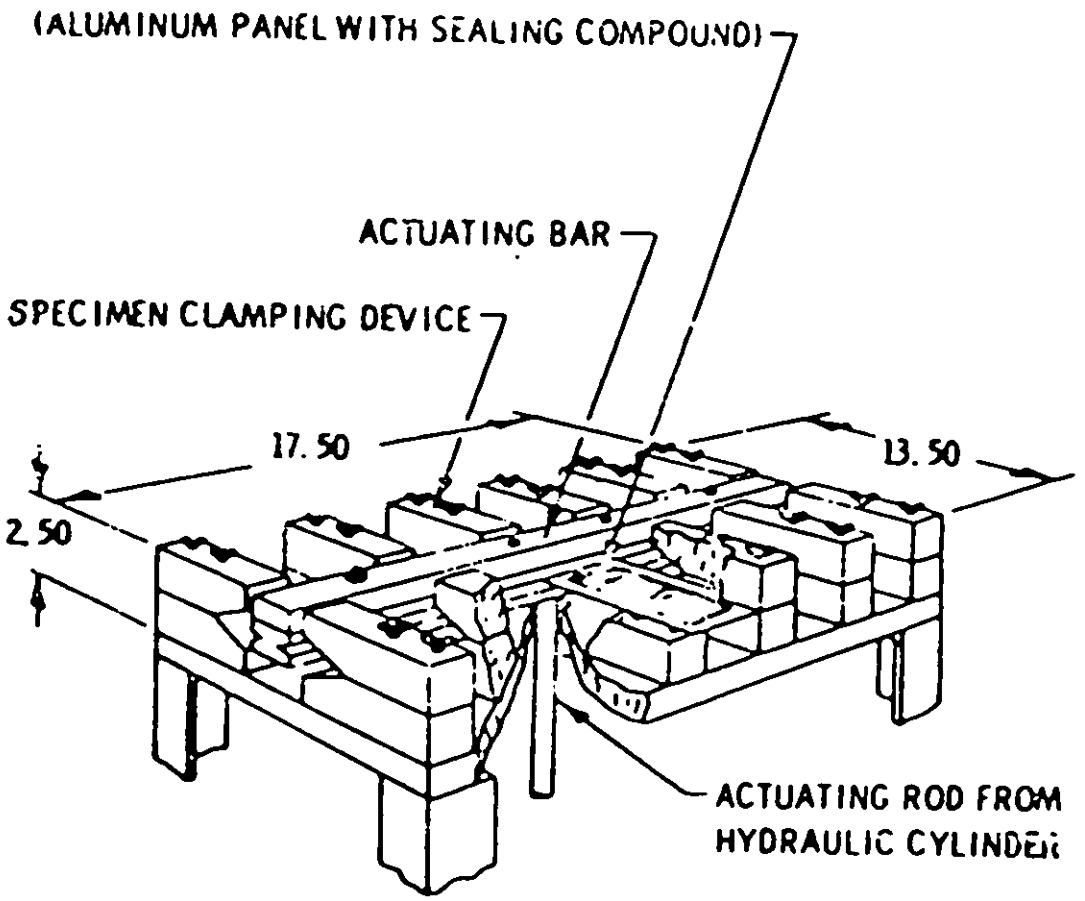


MATERIAL: ALUMINUM ALLOY

DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: DECIMALS ±.010.

FIGURE 1. Flow-test jig

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DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED,
TOLERANCES: ± .015

FIGURE 2. Low temperature flexibility jig

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	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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